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VIRAL AND VIROID DISEASES

Interference with cell-to-cell movement of *Tomato mosaic virus* by transient overexpression of *Arabidopsis thaliana* KELP homologs from different plant species

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Abstract The KELP protein of *Arabidopsis thaliana* (AtKELP), which binds to the *Tomato mosaic virus* (ToMV) movement protein in vitro, can interfere with cell-to-cell movement of the virus under transient overexpression conditions. In this study, we constructed expression plasmids encoding a fluorescent protein fused to AtKELP or to its homolog from *Brassica*, *Nicotiana*, *Solanum*, or *Oryza* species. When expressed transiently, all the AtKELP homologs inhibited the cell-to-cell movement of ToMV with different efficiencies. The results of domain swapping between two selected AtKELP homologs suggest the importance of combination between the N- and C-terminal half regions for high inhibitory function.

Keywords *Tomato mosaic virus* · Movement protein · KELP · Cell-to-cell movement · Inhibitory factor · Transient overexpression

The disease cycle of plant viruses depends on the molecular machinery of the host plants, which means that compatible interactions between virus components and certain host factors are required for virus replication and movement (Lucas 2006; Robaglia and Caranta 2006). That the host range of most plant viruses is limited indicates that other host factors contribute to inducing defense responses against virus attacks or to inhibiting the function of virus components (Ishibashi et al. 2007; Soosaar et al. 2005). To fully understand virus–host interactions at the molecular level, many researchers have recently attempted to identify and characterize the host factors involved in virus infection (Whitham and Wang 2004).

Tomato mosaic virus (ToMV) is a single-stranded RNA virus that belongs to the genus *Tobamovirus* (Ishikawa and Okada 2004). The ToMV genome encodes four proteins from its 5′ to 3′ end: the 126-kDa protein, 180-kDa protein, 30-kDa movement protein (MP), and the coat protein (CP). The 126 and 180-kDa proteins, both of which are translated from genomic RNA, are necessary for virus replication in infected cells. The MP and CP, which are synthesized from a corresponding subgenomic RNA, are required for cell-to-cell movement of the virus through cytoplasmic channels, termed plasmodesmata, and long-distance movement via phloem, respectively. The CP is dispensable for virus cell-to-cell movement; thus, replacement of the CP gene with a reporter gene, such as green fluorescent protein (GFP), allows visualization of the local spread of the virus (Tamai and Meshi 2001).

We previously obtained several cDNAs of host factors that can bind to ToMV MP in vitro, such as protein kinases and transcriptional coactivators (Matsushita et al. 2000, 2001, 2002, 2003; Yoshioka et al. 2004). These factors include the putative transcriptional coactivator KELP of *Arabidopsis thaliana* (AtKELP) (Cormack et al. 1998) and its homolog from *Brassica campestris* (syn. *B. rapa*) (BcKELP). Recently, we have shown that the transient overexpression of AtKELP in *Nicotiana benthamiana* interferes with cell-to-cell movement of ToMV and localization of MP to the plasmodesmata. We have also shown that AtKELP and MP colocalize in cells (Sasaki et al. 2009). These results suggest that overexpressed AtKELP

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can interfere with the cell-to-cell movement of ToMV through its interaction with the MP. AtKELP has also been shown to localize in the nucleus and cytoplasm without virus infection (Sasaki et al. 2009), implying that it may function as a transcriptional coactivator in gene regulation.

Because of the high degree of similarity of BcKELP and AtKELP at the amino acid level (Fig. S1) (Matsushita et al. 2001), we hypothesized that the overexpression of BcKELP in *N. benthamiana* would interfere with MP function and inhibit the cell-to-cell movement of ToMV. To test this hypothesis, we constructed the expression plasmids pART7-AtKELP-DsRed and pART7-BcKELP-DsRed, which encode the respective KELP proteins fused to the *Discosoma* red fluorescent protein (DsRed) (Fig. 1a), by the Gateway cloning method (Invitrogen, Carlsbad, CA, USA). For the construction of pART7-AtKELP-DsRed, a polymerase chain reaction (PCR) was performed with the plasmid pCR-KELP (Sasaki et al. 2009) and primer set attB1/KELP/F (5′-AAAAAGCAGGCTACCATGGAGAAA GAGACGAAG-3′) and attB2/KELP/R (5′-AGAAAGCT GGGTTGACACGCGATTCCATTTT-3′). For the construction of pART7-BcKELP-DsRed, PCR was performed with the plasmid pART7-Bc2, which had been constructed by insertion of the 986-bp *Sma*I–*Xba*I fragment from the plasmid pGEM-Bc2 (Matsushita et al. 2001) between the *Sma*I and *Xba*I sites in pART7, and primer set GW/BcKELP/F01 (5′-AAAAAGCAGGCTACCATGGAGGAAGAAAGC AAG-3′) and attB2/KELP/R. After gel purification of these PCR fragments, a second PCR was performed with the attB1 and attB2 adaptor primers (Sasaki et al. 2009). PCR products with complete attB1 and attB2 recombination sites were cloned into the entry vector pDONR221 (Invitrogen) and transferred to the destination vector pART7-GWC-DsRed (Sasaki et al. 2009) according to the manufacturer's instructions. PCR-derived coding sequences in the entry and destination plasmids were verified by sequencing. Particle bombardment, nuclear staining with 4′,6-diamidino-2-phenylindole (DAPI), and fluorescence observation were performed as described previously (Sasaki et al. 2009).

Leaves of *N. benthamiana* were bombarded with pART7-AtKELP-DsRed or pART7-BcKELP-DsRed and observed with a fluorescence microscope 24 h after bombardment. The red fluorescence of AtKELP-DsRed or BcKELP-DsRed was seen in the cytoplasm and nucleus of the bombarded cells (Fig. 1b). The nuclear localization of these proteins was further confirmed by staining the nucleus with DAPI (Fig. 1b). These results indicated that BcKELP might be a nuclear and cytoplasmic protein, similar to AtKELP.

The leaves were then bombarded with the plasmids pART7-AtKELP-DsRed or pART7-BcKELP-DsRed together with the plasmid piL.erG3, which contained a modified ToMV genome that encoded an endoplasmic reticulum-targeted GFP instead of the CP (Tamai and Meshi 2001). Because the modified ToMV produced a nondiffusive GFP in a replication-dependent manner, monitoring the distribution of GFP in leaf tissues allowed us to examine whether the virus moved from the initially bombarded cell to neighboring cells (Fig. 1c). As a control, we used the plasmid pART7-DsRed, which encoded free DsRed. To assess the extent of virus movement, we examined the infection sites, where both GFP and DsRed were found in the initially bombarded single cells. Some infection sites encompassed multiple neighboring cells with GFP around the initially bombarded cells, whereas other infection sites did not. Throughout this study, we examined at least 30 infection sites in each bombardment experiment and determined the percentages of infection sites with multiple cells having GFP (number of infection

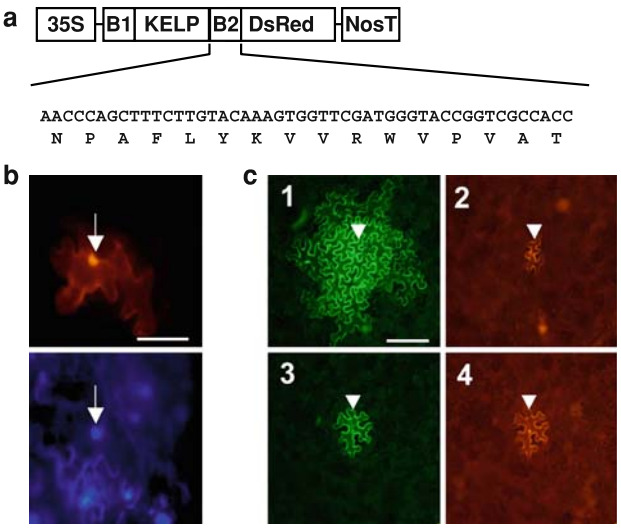


Fig. 1 Diagram of an expression cassette for KELP-DsRed fusion proteins and representative images of leaves of *Nicotiana benthamiana* after bombardment. **a** Each expression cassette contains the *Cauliflower mosaic virus* 35S promoter (35S), the Gateway attB1 (B1) and attB2 (B2) sequences, a *KELP* cDNA (KELP), the *DsRed* cDNA (DsRed), and the nopaline synthase terminator (NosT). An extra amino acid sequence in a KELP-DsRed fusion protein is shown below the cassette. **b** Nuclear and cytoplasmic localization of BcKELP-DsRed in an epidermal cell. Leaves of *N. benthamiana* at 24 h after bombardment with pART7-BcKELP-DsRed were stained with DAPI and observed with an epifluorescence microscope with a DsRed filter unit (*upper*) and a DAPI filter unit (*lower*). *Arrows* point to the position of the nucleus. **c** GFP-expressing infection sites consisting of multiple cells (*1* and *2*) or a single cell (*3* and *4*) 48 h after bombardment with piL.erG3L and pART7-BcKELP-DsRed. A multiple-cell infection site where GFP fluorescence was distributed in cells surrounding an initially bombarded cell indicated successful cell-to-cell movement of the virus. In contrast, a single-cell infection site where GFP fluorescence was confined to the initially bombarded cell indicated inhibition of virus movement. Images were taken of the same field with filter units for GFP (*left*) and DsRed (*right*). Initially bombarded cells are indicated by *arrowheads*. *Scale bars* **b** 100 μm, **c** 200 μm

sites with multiple cells divided by total number of infection sites times 100). We found that the average percentage of multiple-cell infection sites cobombarded with piL.erG3 and AtKELP-DsRed, BcKELP-DsRed, or free DsRed was approximately 24, 6, or 95%, respectively (Fig. 2). These results indicated that BcKELP inhibited the cell-to-cell movement of ToMV even more effectively than did AtKELP.

Using public databases, we found other AtKELP homologs from several plant species such as *N. tabacum* (accession no. EB681766), *Solanum lycopersicum* (accession no. BM411434), and *Oryza sativa* (accession no. NM_001050549). Our results encouraged us to investigate whether the AtKELP homologs of *N. tabacum* (NtKELP), *S. lycopersicum* (SlKELP), and *O. sativa* (OsKELP) were able to interfere with the cell-to-cell movement of ToMV. We extracted total RNA from *N. tabacum* cv. Samsun NN, *S. lycopersicum* cv. Micro-Tom, and *O. sativa* cv. Nipponbare by using the Concert Plant RNA Reagent (Invitrogen) or RNeasy Plant Mini Kit (Qiagen, Valencia, CA, USA) and synthesized first strand cDNA from the total RNA extract, an oligo-dT primer, and ReverTra Ace reverse transcriptase (Toyobo, Osaka, Japan). The cDNAs of *NtKELP*, *SlKELP*, and *OsKELP* were amplified by PCR with their respective primer sets of GW/NtKELP/F01 (5′-AAAAAGCAGGCTACCATGGATTCAGAAACTTCG-3′) and GW/NtKELP/R01 (5′-AGAAAGCTGGGTTACACCTTGACTCCATTTT-3′); GW/SlKELP/F01 (5′-AAAAAGCAGGCTACCATGGATTCTGAAACTTCC-3′) and GW/SlKELP/R01 (5′-AGAAAGCTGGGTTAGCCTTCGACTCCATTTT-3′); and GW/OsKELP/F01 (5′-AAAAAGCAGGCTACCATGGACGAGGAGACGCAG-3′) and GW/OsKELP/R01 (5′-AGAAAGCTGGGTTGTCCGATGATTCTCCAAG-3′). The subsequent cloning of their cDNAs to the pDONR221 and pART7-GWC-DsRed vectors was performed as described for AtKELP and BcKELP. Finally, we obtained three expression plasmids, pART7-NtKELP-DsRed, pART7-SlKELP-DsRed, and pART7-OsKELP-DsRed, which encoded DsRed-fused NtKELP, SlKELP, and OsKELP, respectively (Fig. 1a). The comparison of the cloned cDNA sequences with those in public databases revealed that the *NtKELP* sequences contained two variations in the nucleotide sequence and consequently in the amino acid sequence (Fig. S1).

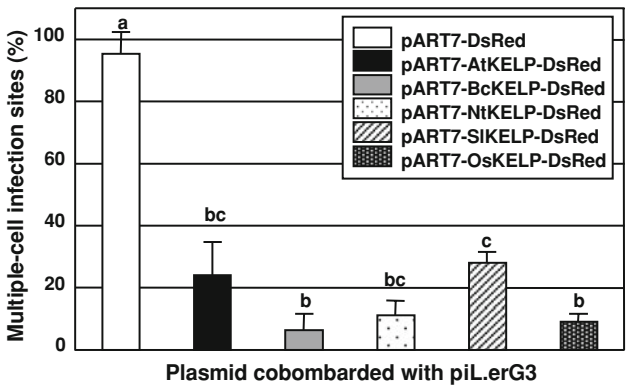


Fig. 2 Inhibitory effects of transient expression of KELP homologs on virus cell-to-cell movement in *Nicotiana benthamiana* cobombarded with piL.erG3L and either pART7-DsRed, pART7-AtKELP-DsRed, pART7-BcKELP-DsRed, pART7-NtKELP-DsRed, pART7-SlKELP-DsRed, or pART7-OsKELP-DsRed. The number of GFP-expressing infection sites consisting of a single cell or multiple (more than two) cells that also contained a cell with DsRed fluorescence was determined 48 h after bombardment. For each bombardment, at least 30 infection sites were examined and the percentage of multiple-cell infection sites was calculated (number of multiple-cell infection sites divided by total number of infection sites times 100). *Bar graphs* illustrate the averaged percentages of multiple-cell infection sites from two (for pART7-DsRed) or three (for the others) independent experiments. *Error bars* indicate standard deviations. *Different letters* (*a*, *b* and *c*) above the *bars* indicate significant differences at $P < 0.05$ by Tukey's test

Bombardment of *N. benthamiana* with the expression plasmids for the KELP homologs without piL.erG3 and subcellular localization analyses by DsRed observation and DAPI staining showed that NtKELP-DsRed, SlKELP-DsRed, and OsKELP-DsRed were localized in the nucleus and the cytoplasm, similar to AtKELP and BcKELP (Fig. S2), suggesting that these AtKELP homologs might function in a similar way to AtKELP. In fact, when pART7-NtKELP-DsRed, pART7-SlKELP-DsRed, or pART7-OsKELP-DsRed was bombarded with piL.erG3, the average percentages of multiple-cell infection sites were approximately 11, 28, or 9%, respectively (Fig. 2). Thus, our results indicated that, when transiently overexpressed, not only AtKELP and BcKELP but also NtKELP, SlKELP, and OsKELP could inhibit the cell-to-cell movement of ToMV. Because the degree of the inhibition varied among these homologs, we compared their differences statistically using an online program for Tukey's test (http://www.gen-info.osaka-u.ac.jp/testdocs/tomocom//tukey.html) (Fig. 2). The statistical analysis suggested that the inhibitory effects of BcKELP and OsKELP were significantly higher than that of SlKELP, whereas AtKELP and NtKELP might have intermediate effects among them.

The amino acid sequences in the N-terminal half are relatively diverse among KELP proteins, whereas those in the C-terminal half are highly conserved (Fig. S1) (Matsushita et al. 2001). Our previous in vitro studies demonstrated that BcKELP contained an MP-binding domain in the N-terminal half (amino acids 1–86) (Matsushita et al. 2001). Thus, it was possible that the N-terminal half of the KELP proteins might determine the degree of their inhibitory effect. To test this possibility, we selected BcKELP and SlKELP to generate chimeras, designated KELP(BS) and KELP(SB). The KELP(BS) chimera consisted of the

N-terminal half (amino acids 1–86) of BcKELP and the C-terminal half (amino acids 97–175) of SlKELP. KELP(SB) consisted of the N-terminal half (amino acids 1–96) of SlKELP and the C-terminal half (amino acids 87–165) of SlKELP (Fig. S1). We constructed the plasmids pART7-KELP(BS)-DsRed and pART7-KELP(SB)-DsRed for transient expression of DsRed-fused KELP(BS) and KELP (SB), respectively, as follows: DNA fragments containing the 5′-half and 3′-half *BcKELP* sequences were amplified by PCR with pART7-BcKELP-DsRed as a template, in combination with primer sets KELP(BS)/F01 (5′-AAAAA GCAGGCTACCATGGAGGAAGAAAGCAAGGC-3′) and KELP(BS)/R01 (5′-TCGTCATCATACTCCTTATCTCCCT CCTTAGCTCTTTCTT-3′); and KELP(SB)/F02 (5′-GCAA AGTGAAAAGCAGCGGTAACAAAGAGTTTGACGAT GA-3′) and KELP(SB)/R02 (5′-AGAAAGCTGGGTTGAC ACGCGATTCCATTT-3′), respectively. DNA fragments containing the 5′-half and 3′-half *SlKELP* sequences were obtained using pART7-SlKELP-DsRed as a template, in combination with primer sets KELP(SB)/F01 (5′-AAA AAGCAGGCTACCATGGATTCTGAAACTTCCAA-3′) and KELP(SB)/R01 (5′-TCATCGTCAAACTCTTTGTT ACCGCTGCTTTTCACTTTGC-3′); and KELP(BS)/F02 (5′-AAGAAAGAGCTAAGGAGGGAGATAAGGAGTA TGATGACGA-3′) and KELP(BS)/R02 (5′-AGAAAGCT GGGTTAGCCTTCGACTCCATTT-3′), respectively. A second round of PCR for each chimeric cDNA was performed with the attB1 and attB2 adaptor primers to combine two appropriate fragments through an overlapping sequence at the junction. The final chimeric cDNAs were recombined into pDONR221 and pART7-GWC-DsRed vectors as described.

N. benthamiana was bombarded with pART7-KELP (BS)-DsRed or pART7-KELP(SB)-DsRed together with piL.erG3. For comparison, pART7-BcKELP-DsRed or pART7-SlKELP-DsRed was also used. Forty-eight hours after bombardment, the percentages of multiple-cell infection sites with BcKELP-DsRed, SlKELP-DsRed, KELP (BS)-DsRed, and KELP(SB)-DsRed were approximately 2, 23, 14, and 20%, respectively (Fig. 3). This result, in addition to statistical analysis, suggested that domain swapping might tend to decrease the inhibitory effect of BcKELP-DsRed but not of SlKELP-DsRed on virus cell-to-cell movement. Thus, the N- and C-terminal halves of BcKELP could be functionally replaced by the corresponding regions of SlKELP, although both of the inherent N- and C-terminal halves of BcKELP might be required for its full ability to inhibit virus movement. Our previous finding that the overexpression of an AtKELP derivative lacking the C-terminal half blocked the cell-to-cell movement of ToMV less effectively than that of the intact AtKELP (Sasaki et al. 2009) may also suggest a cooperative role between the N- and the C-terminal halves in KELP function.

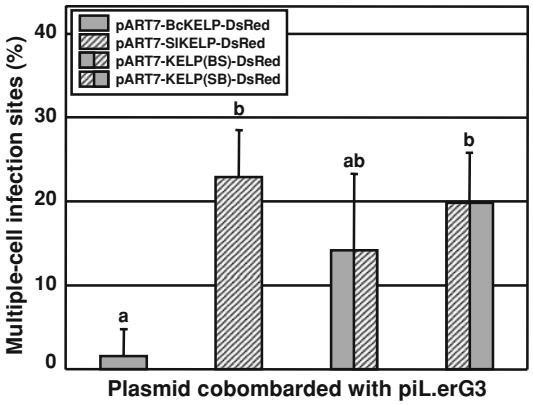


Fig. 3 Inhibitory effects of transient expression of chimeric BcKELP and SlKELP on virus cell-to-cell movement in *Nicotiana benthamiana* at 48 h after cobombardment with piL.erG3L and either pART7-BcKELP-DsRed, pART7-SlKELP-DsRed, pART7-KELP(BS)-DsRed, or pART7-KELP(SB)-DsRed. After each bombardment, we examined at least 30 infection sites. The average percentage of multiple-cell infection sites was calculated as described in the legend of Fig. 2. *Bar graphs* illustrate the average percentages of multiple-cell infection sites from four independent experiments for all plasmids. *Error bars* indicate standard deviations. *Different letters* (*a* and *b*) above the *bars* indicate significant differences at $P < 0.05$ by Tukey's test

In conclusion, the results of this study show that AtKELP, another MP-binding protein BcKELP, and three other KELP proteins can inhibit the cell-to-cell movement of ToMV, at least when overexpressed. These results, in addition to the amino acid sequence homology to AtKELP and BcKELP, suggest that NtKELP, SlKELP, and OsKELP may be MP-binding proteins. Previously, we found that overexpressed AtKELP colocalized with ToMV MP and disturbed the targeting of the MP to plasmodesmata (Sasaki et al. 2009). On the basis of these findings, we propose that MP function may be hampered through interactions with the KELP proteins, although actual interactions in planta have not been proved. On the other hand, these KELP proteins are assumed to function in host gene expression as transcriptional coactivators because their C-terminal halves contain the single-stranded DNA binding motifs conserved in other transcriptional coactivators (Fig. S1) (Cormack et al. 1998). Although the C-terminal half of AtKELP is not absolutely required to disrupt the cell-to-cell movement of ToMV (Sasaki et al. 2009), the overexpression of the full-size KELP proteins may possibly stimulate unknown host defense responses to virus infection.

Our results show that the ability of the KELP proteins to inhibit movement of the virus is widely conserved among various plant species. *B. campestris* and *O. sativa* are possibly nonhosts for ToMV, based on the results of infectivity assays of the most closely related *Tobacco mosaic virus* (Okada et al. 1988; Zhang et al. 1999). In contrast, *A. thaliana*, *N. tabacum*, and *S. lycopersicum* are

all systemic hosts for the virus (Ohshima et al. 1998; Weber et al. 1992). Thus, these KELP proteins may not be major determinants for the host specificity of ToMV but may more or less affect the efficiency of virus cell-to-cell movement in natural conditions. In this study, we also show that the degree of the inhibitory effect differed among the KELP proteins. Further investigation will be necessary to determine whether such differences are attributable to qualitative or quantitative differences of the KELP proteins in their binding affinity to MP, protein stability or expression level. From the view of plant protection from viruses, we expect that we may be able to exploit the *KELP* genes and their inhibitory mechanisms to generate novel virus-resistant plants.

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